Xun Xu

Education Experience

2006.9 - 2010.6 **BEng in Automation**, *Sichuan University*, China

2010.9 - 2012.6 **MSc in Control Theory**, *Sichuan University*, China

2012.9-2016.12 **PhD in Computer Science**, *Queen Mary University of London*, UK

PhD Thesis

Title *Semantic Spaces for Human Behaviour Analysis*

Supervisors Prof. Shaogang Gong, Prof. Tao Xiang and Dr. Timothy Hospedales

Working Experience

2023.06 - Now **Senior Scientist II**, *Infocomm Research (I2R), A-STAR, Singapore*

2022.04 - 2023.06 **Scientist III**, *Infocomm Research (I2R), A-STAR, Singapore*

2019.12 - 2022.03 **Scientist II**, *Infocomm Research (I2R), A-STAR, Singapore*

Computer vision research towards fewer labelled data

2016.09 - 2019.11 **Research Fellow**, *National University of Singapore, Singapore*

Develop motion segmentation algorithm for autonomous driving and deep learning for general multi-model fitting

2015.09 - 2016.06 **Consultant**, *Vision Semantics Ltd.*, London

Person detection proposal

Programming Skills

Daily Use Matlab, Python, PyTorch

Familiar with C/C++, Tensorflow, PHP, Javascript, html

Selected Publications

# indicates equal contribution.

* indicates corresponding author.

Book Chapter

1. **X. Xu**, S. Gong and T. Hospedales. *Zero-Shot Crowd Behaviour Recognition*, In Murino, Shah, Cristani, Savarese (Eds.), Group and Crowd Behaviour Understanding in Computer Vision, Elsevier, 2016.

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GoogleScholar: Xun Xu • LinkedIn: xu-xun • GitHub: alex-xun-xu

Last Updated on 25th Jan 2024

Journal

2. Y. Su#, **X. Xu**#*, T. Li and K. Jia*, Revisiting Realistic Test-Time Training: Sequential Inference and Adaptation by Anchored Clustering Regularized Self-Training, IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2024
3. J. Liao, **X. Xu***, M. C. Nguyen, A. Goodge and C. S. Foo, COFT-AD: COntrastive Fine-Tuning for Few-Shot Anomaly Detection, IEEE Transactions on Image Processing (**TIP**), 2024
4. **X. Xu**, J. Liao, L. Cai, M. C. Nguyen, K. Lu, W. Zhang, Y. Yazici and C. S. Foo, Revisiting Pretraining for Semi-Supervised Learning in the Low-Label Regime, **Neurocomputing**, 2023
5. Y. Su#, **X. Xu**#* and K. Jia*, Weakly Supervised 3D Point Cloud Segmentation via Multi-Prototype Learning, IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**), 2023
6. N. Liu, **Xun Xu***, T. Celik, Z. Gan and H. C. Li*, Transformation-Invariant Network for Few-Shot Object Detection in Remote Sensing Images, IEEE Transactions on Geoscience and Remote Sensing (**TGRS**), 2023
7. C. K. Joshi, F. Liu, **X. Xu**, J. Lin and C. S. Foo, On Representation Knowledge Distillation for Graph Neural Networks, IEEE Transactions on Neural Network and Learning Systems (**TNNLS**), 2023
8. **X. Xu**, C. M. Nguyen, Y. Yazici, K. Lu, H. Min and C. S. Foo, SemiCurv: Semi-Supervised Learning for Curvilinear Structure Segmentation, IEEE Transactions on Image Processing (**TIP**), 2022
9. Z. Jiang#, **X. Xu#***, C. Zhang, C.S. Foo and C. Zhu*, MAGNet: a Multi-Attention Generative Adversarial Network for Defocus Blur Detection, IEEE Transactions on Image Processing (**TIP**), 2022
10. C. M. Nguyen, L. Zhang, **X. Xu**, K. Lu, B. Unnikrishnan and C. S. Foo, Semi-supervised Regression with Multi-view Diversity and Consistency, Machine Learning (**ML**) 2022
11. L. Cai, **X. Xu**, L. Zhang and C.S. Foo, Exploring Spatial Diversity for Region-based Active Learning, IEEE Transactions on Image Processing (**TIP**), 2021
12. **X. Xu,** L.F. Cheong, Z. Li and C. Zhu, Learning Clustering for Motion Segmentation, IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**), 2021
13. C. Zhang, C. Zhu and **X. Xu**, Revisiting Image Ordinal Estimation: How to Deal with Ordinal Relationship in Deep Learning?, Journal of Electronic Imaging, 2019

14. **X. Xu**, L.F. Cheong, Z. Li, 3D Rigid Motion Segmentation with Mixed and Unknown Number of Models, IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2019.
15. C. Zhang, **X. Xu**, C. Zhu, Image Ordinal Classification with Deep Multi-View Learning, IET Electronics Letters, 2018.
16. C. Zhang, C. Zhu, **X. Xu**, Y. Liu, J. Xiao and T. Tillo, Visual aesthetic understanding: Sample-specific aesthetic classification and deep activation map visualization, Signal Processing: Image Communication, 2018
17. **X. Xu**, T. Hospedales, S. Gong, Zero-Shot Action Recognition by Word-Vector Embedding, International Journal of Computer Vision (**IJCV**), 2017.
18. **X. Xu**, T. Hospedales, S. Gong, Discovery of Shared Semantic Spaces for Multi-Scene Video Query and Summarization, IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**), 2017.

Conference

19. Y. Su, **X. Xu*** and K. Jia*, Towards Real-World Test-Time Adaptation: Tri-Net Self-Training with Balanced Normalization, AAAI Conference on Artificial Intelligence (**AAAI**) 2024
20. R. Pang, H. Tan, Y. Yang, **X. Xu**, N. Liu and P. ZhangA Novel SegNet model for Crack Image Semantic Segmentation in Bridge Inspection, Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**) 2024 (Oral Presentation)
21. Y. Li, **X. Xu***, Y. Su and K. Jia*, On the Robustness of Open-World Test-Time Training: Self-Training with Dynamic Prototype Expansion, IEEE/CVF Conference on Computer Vision (**ICCV**) 2023 (**Oral Presentation**; 1.8% acceptance rate)
22. K. Lu, M. C. Nguyen, **X. Xu** and C. S. Foo, On Adversarial Robustness of Audio Classifiers, International Conference on Acoustics, Speech, & Signal Processing (**ICCASP)**, 2023
23. Y. Su#, **X. Xu#***, K. Jia*, Revisiting Test-Time Training: Sequential Inference and Adaptation by Anchored Clustering, Neural Information Processing Systems (**NeurIPS**), 2022
24. X. Shi, **X. Xu***, W. Zhang, X. Zhu, K. Chen, C. S. Foo and K. Jia, Open-Set Semi-Supervised Learning for 3D Point Cloud Understanding, IEEE Conference on Pattern Recognition (**ICPR**), 2022 (Oral Presentation)

25. L. Cai, R. S. Pahwa, **X. Xu**, J. Wang, L. Zhang and C. S. Foo, Exploring Active Learning for Semiconductor Defect Segmentation, IEEE Conference on Image Processing (**ICIP**), 2022

26. J. Wang, R. Chang, **X. Xu**, L. Cai, C. S. Foo and R. S. Pahwa, Improved Bump Detection and Defect Identification for HBMs using Refined Machine Learning Approach, IEEE 24th Electronics Packaging Technology Conference (**EPTC**), 2022 (Best Academic Paper Award)

27. W. Zhang, **X. Xu***, F. Liu, L. Zhang, C. S. Foo, On Automatic Data Augmentation for 3D Point Clouds, British Machine Vision Conference (**BMVC**), 2021

28. K. Lu, C. M. Nguyen, **X. Xu**, K. Chari, Y. J. Goh and C. S. Foo, ARMOURED: Adversarially Robust MOdels using Unlabeled data by REgularizing Diversity, International Conference on Learning Representations (**ICLR**), 2021

29. S. Deng#, **X. Xu**#, C. Wu, K. Chen, K. Jia, 3D AffordanceNet: A Benchmark for Visual Object Affordance Understanding, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2021

30. L. Cai, **X. Xu**, J. H. Liew, C. S. Foo, Revisiting Superpixels for Active Learning in Semantic Segmentation with Realistic Annotation Costs, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2021

31. Z. Jiang, **X. Xu**, C. Zhang, X. Liu and C. Zhu, MultiANet: a Multi-Attention Network for Defocus Blur Detection, IEEE International Workshop on Multimedia Signal Processing (**MMSP**), 2020 (Oral Presentation)

32. **X. Xu**, G.H. Lee, Weakly Supervised Semantic Point Cloud Segmentation: Towards 10X Fewer Labels, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2020

33. C. Zhang, S. Liu, **X. Xu** and C. Zhu, C3AE: Exploring the Limits of Compact Model for Age Estimation, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2019

34. K. Lin, N. Jiang, L.F. Cheong, J. Lu and **X. Xu**, Robust Video Background Identification by Dominant Rigid Motion Estimation, Asian Conference on Computer Vision (**ACCV**), 2018

35. **X. Xu**, L.F. Cheong and Z. Li, Motion Segmentation by Exploiting Complementary Geometric Models, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2018

36. C. Zhang, C. Zhu, J. Xiao, **X. Xu*** and Y. Liu, Image Ordinal Classification and Understanding: Grid Dropout with Masking Label, IEEE International Conference on Multimedia and Expo (**ICME**), 2018 (Oral Presentation)

37. **X. Xu**, T. Hospedales, and S. Gong. Multi-Task Zero-Shot Action Recognition with Prioritised Data Augmentation, European Conference on Computer Vision (**ECCV**), 2016
38. **X. Xu**, T. Hospedales, S. Gong, Semantic Embedding Space for Zero-Shot Action Recognition, International Conference on Image Processing (**ICIP**), 2015 (**Oral, Top 10% paper**)
39. **X. Xu**, S. Gong, T. Hospedales, Cross-Domain Traffic Scene Understanding by Motion Model Transfer, ACM International Conference on Multimedia (ACM MM), Workshop on Analysis and Retrieval of Tracked Events and Motion in Imagery Streams, 2013
40. **X. Xu**, Y. Liu, MEMS based Human-Machine Interaction System for virtual touring, International Conference on Computer Science and Network Technology (ICCSNT), 2011

Projects

Title: Towards Realistic Deep Learning for 3D Vision
Duration: Aug. 2023 – Jul. 2026
Grant Name: MTC Programmatic Funds
Funding Agency: A*STAR, Singapore
Amount: SGD $9.77M (SGD $1.1M allocated to Co-I)
Role: **Co-I**

Title: Learning with Less Data
Duration: Apr. 2021 – Mar. 2024
Grant Name: AME Programmatic Funds
Funding Agency: A*STAR, Singapore
Amount: SGD $8.94M (SGD $1.4M allocated to Co-I)
Role: **Co-I**

Title: Exploiting Unlabeled Data, Cheaper Labels and Efficient Annotation for 3D Point Cloud Deep Learning (~**25% acceptance**)
Duration: Apr. 2021 - Apr. 2024
Grant Name: Career Development Award (CDA)
Funding Agency: A*STAR, Singapore
Amount: SGD $238,000

Role: **PI**

Professional Activities

Members

IEEE Senior Member

Area Chair

ACM Multimedia 2024

Reviewer

Journals

International Journal of Computer Vision (IJCV)
IEEE Transactions on Neural Network and Learning Systems (TNNLS)
IEEE Transactions on Image Processing (TIP)
IEEE Transactions on Multimedia (TMM)
IEEE Transactions on Systems, Man and Cybernetics: Systems (TSMC)
IEEE Transactions on Cybernetics (TOC)
IEEE Transactions on Circuits System and Video Technology (TCSVT)
ACM Transactions on Knowledge Discovery from Data (TKDD)
Pattern Recognition (PR)
Transactions on Machine Learning Research (TMLR)

Conferences

Conference on Neural Information Processing Systems (NeurIPS)
International Conference on Machine Learning (ICML)
International Conference on Learning Representations (ICLR)
IEEE Conference on Computer Vision and Pattern Recognition (CVPR)
European Conference on Computer Vision (ECCV)
IEEE International Conference on Computer Vision (ICCV)
AAAI conference on Artificial Intelligence (AAAI)
ACM Multimedia Conference (ACM MM)
British Machine Vision Conference (BMVC)
International Conference on Robotics and Automation (ICRA)
International Conference on Intelligent Robots and Systems (IROS)

Invited Talks

2022 International Conference on Digital Society and Intelligent Systems – Dec 2022
Southwest Jiaotong University – 7 Mar 2022
A*STAR – 9 Dec, 2019
Tongji University – 18 May, 2019
University of Electronic Science and Technology of China – 23rd, Nov, 2018
Soochow University - 10th Apr, 2018
Vision And Learning SEminer (VALSE) - 9th Aug, 2017
Sichuan University - 26th, Dec, 2017

University of Electronic Science and Technology of China - 27th, Dec, 2017

Awards

EPTC 2022 Best Academic Paper Award (SGD$1500 shared by co-authors) 2023
A*STAR Niche Skill Allowance (SGD$1600/month) 2021
Queen Mary Postgraduate Research Fund (£550) 2016
Top 10% paper in ICIP 2015
IEEE Signal Processing Society Travel Grant for ICIP ($600) 2015
QMUL Intelligent Sensing Summer School Challenge (3rd Prize) 2013

Students

PhD

Chengxin Liu, Sep 2023 – Now (HUST, university funded visiting student)

Yongyi Su, Sep 2021 – Now (SCUT, CSC funded visiting student)

Nanqing Liu, Jan 2023 – Now (Southwest Jiaotong University, CSC funded visiting student)

Rong Pang, Apr 2023 – Now (Southwest Jiaotong University, China Merchant's Group funded visiting student)

Jianning Deng, Sep 2022 – May 2023 (University of Edinburgh, co-supervised with Prof. Chris Lu, ARAP scholarship)

Chao Zhang, Sep 2014 – 2019 (PhD@UESTC, co-supervised with Prof. Ce Zhu)

Master

Junchao Wang, Aug 2023 – Now (Master@NUS, visiting student)

Jinpeng Lin, Sep 2022 – Now (Master@SWJTU, co-supervised with Prof. Tianrui Li)

Yushu Li, Sep 2022 – Now (Master@SCUT, SIPGA funded visiting student)

Yijin Chen, Sep 2021 – Now (Master@SCUT, co-supervised with Prof. Kui Jia)

Xian Shi, Sep 2020 – Jun 2023 (Master@SCUT, co-supervised with Prof. Kui Jia)

Zeyu Jiang, Sep 2019 – Jun 2022 (Master@UESTC, co-supervised with Prof. Ce Zhu)

Qian Song, Aug 2017 - Apr 2018 (Master@NUS, co-supervised with Prof. Cheong Loong-Fah)

References

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Dr. Timothy Hospedales,
University of Edinburgh,
E-mail: t.hospedales@ed.ac.uk

Prof. Ce Zhu, FIEEE
University of Electronic and Science Technology of China,
E-mail: eczhu@uestc.edu.cn

Assoc. Prof. Gim Hee Lee,
National University of Singapore,
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